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#ATSU '18
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Kwame Nkrumah University of Science and Technology
College of Engineering

COE/EE 152: Basic Electronics

Mid-Semester Examination

2013/2014 Academic Year

$1\frac{1}{2}$ Hours

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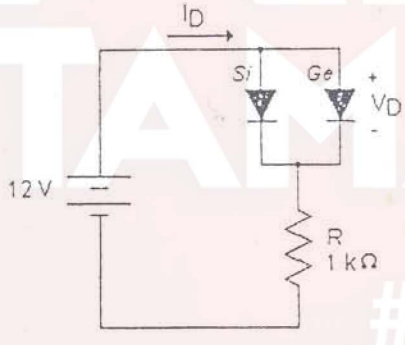
NOTE:
If a diode model is not stated assume CVD model with knee voltages of 0.7 V and 0.3 V for silicon and germanium repectively. Some diode are however ideal and can be easily identified from the question
ALL Questions carry equal marks

MULTIPLE CHOICE. Choose the one alternative that best completes the statement or answers the question and write the corre answer in the space provided

1) Reverse leakage current in a p-n junction diode is caused by
A) majority carriers in the n-type
B) majority carriers in the p- and n- type
C) minority carriers in the p-type
D) minority carriers in the p- and n- type

1) D

2) The value of V_D in this circuit is _____.



A) 11.3 V B) 0.3 V C) 10.6 V D) 0.7 V

2) B

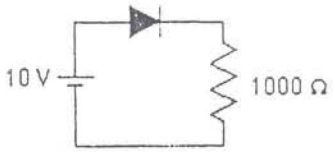
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ATSU'S INITIATIVE

3) A half-wave rectifier circuit consists of a diode(D_1) with an R_{D1} of 10 Ω, a 200 μF filter capacitor, and an R_L of 2 kΩ. How much longer does it take the capacitor to discharge than charge? 3) C

A) 100 times as long
B) 400 times as long
C) 200 times as long
D) 1,000 times as long

4) For this circuit, determine the load-line intersection with the two axis. 4) B



A) V_D = 1 V and I_D = 1 mA
B) V_D = 10 V and I_D = 10 mA
C) V_D = 10 V and I_D = 1 mA
D) V_D = 1 V and I_D = 10 mA

5) Pentavalent atoms are often referred to as _____. 5) C

A) majority carriers B) minority carriers C) donor atoms D) accepter atoms

6) When a *p-n* junction is reverse-biased, the depletion layer is _____ and the device acts as a near-perfect _____. 6) C

A) narrowed; conductor
B) narrowed; insulator
C) widened; insulator
D) widened; conductor

7) In the Zener region the current _____ and the voltage across the diode _____. 7) B

A) can increase a lot; can increase a lot
B) can increase a lot; is almost constant
C) is almost constant; can increase a lot
D) is almost constant; is almost constant

8) A *p-n* junction is forward biased when _____. 8) B

A) the applied potential causes the *n*-type material to be more positive than the *p*-type material
B) the applied potential causes the *n*-type material to be more negative than the *p*-type material
C) both materials are at the same potential
D) None of these

9) A *p-n* junction is reverse biased when _____. 9) 9

A) the current flow across the junction is based on minority carrier transfer
B) the applied potential causes the *n*-type material to be more positive than the *p*-type material
C) the applied potential causes the *p*-type material to be more negative than the *n*-type material
D) All of the above

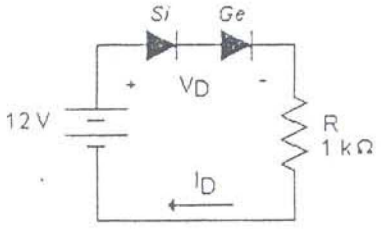
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10) The number of holes is equal to the number of electrons and electron-hole pair generation increases greatly a: temperature rises in: 10) A

A) intrinsic semiconductors
B) p-type semiconductors
C) n-type semiconducotors
D) conducotors

11) The resistor voltage and resistor current in this circuit are _____. 11) D



A) 10 V, 5 mA
B) 11 V, 2 mA
C) 2 V, 11 mA
D) 11 V, 11 mA

12) The _____ diode is a short circuit for the region of conduction and it is an open circuit in the region of nonconduction. 12) A

A) ideal
B) power
C) small-signal
D) typical

13) One electron volt is equal to _____ Joules. 13) C

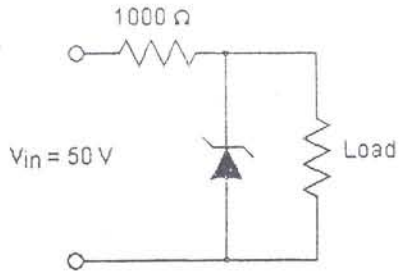
A) 3.6×10^{-16}
B) 0.7×10^{-16}
C) 1.6×10^{-19}
D) It depends on which shell the electron is in

14) When silicon is doped with either a pentavalent or a trivalent impurity its resistance will _____. 14) B

A) make the resistance stable against variation due to temperature
B) decrease
C) increase
D) None of the above

BELIEVE IN THE PROCESS

15) What are the minimum and maximum values of current flowing in the variable load resistor while the diode i operating in the Zener region? The zener voltage is 10 V. 15) A



A) 8 mA and 40 mA
B) 8 mA and 35 mA
C) 12.5 mA and 40 mA
D) Need to know the load resistance to determine the values.

16) The point of intersection between the characteristic curve of the diode and the resistors loadline is known as t 16) D
_____.
A) point of operation B) quiescent point C) Q-point D) All of the above

17) If one silicon diode and one germanium diode are connected in series, the voltage drop across the combinatio 17) B
the two diodes will be equal to _____.
A) the forward drop equal to that of the silicon diode
B) the forward drop equal to that of the sum of the voltage drops across the two diodes
C) the forward drop equal to that of the difference of the voltage drops across the two diodes
D) the forward drop equal to that of the germanium diode

18) A Zener diode is designed to operate in the _____ region of its characteristic curve. 18) B
A) reverse bias B) reverse breakdown
C) zero voltage D) forward operating

19) A power supply filter 19) C
A) increases V_{ave} B) changes dc to ac
C) decreases ripple voltage D) increases ripple voltage

20) Increasing the temperature of a forward-biased diode _____. 20) C
A) causes forward current to decrease
B) has no significant effect effect on the forward current
C) causes forward current to increase
D) None of these

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21) All the following are properties of semiconductors except
A) Has a crystalline structure
B) resistance is inversely proportions
C) Conducts in only one direction
D) Electron emissioin

21) B

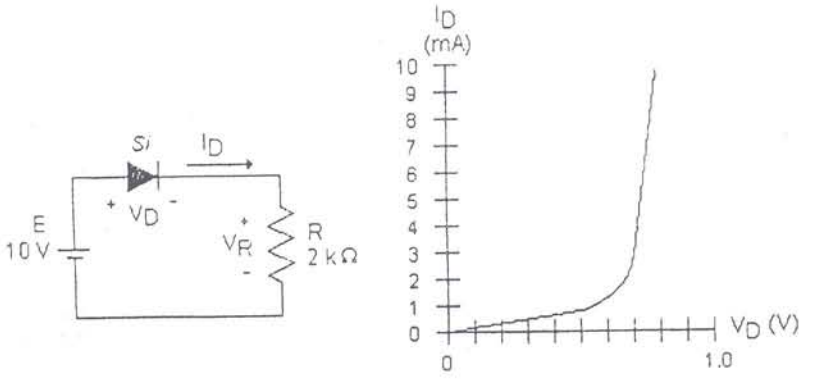
22) The act of giving off light by applying an electrical source of energy is called _____.
A) laser
B) photons
C) light power
D) electroluminescence

22) D

23) With regard to peak inverse voltage for a half-wave rectifier, which formula is correct?
A) PIV ≅ $3V_{s(pk)}$ B) PIV ≅ $V_{s(pk)}$ C) PIV ≅ $4V_{s(pk)}$ D) PIV ≅ $2V_{s(pk)}$

23) B

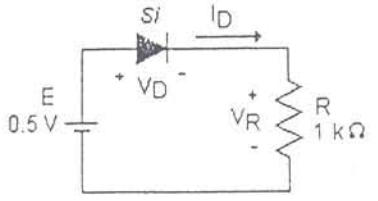
24) For this series diode configuration, use the diode characteristic to estimate the value of V_R.



A) 92 mV B) 9.2 V C) 0.92 V D) 10 V

24) B

25) The practical value of the current I_R in this circuit is _____.



A) 0.5 A B) 0.5 mA C) 0 A D) 5 mA

25) C

26) When a 60 Hz sinusoidal signal voltage is applied to the input of a half-wave rectifier, the output frequency is
A) 120 Hz. B) 90 Hz. C) 60 Hz. D) 30 Hz.

26) C

27) If input frequency is 60 Hz, the output frequency of a bridge rectifier is 27) A
A) 120 Hz. B) 60 Hz. C) 240 Hz. D) 30 Hz.

28) A silicon diode has a voltage to ground of 117 V from the anode. The voltage to ground from the cathode is 117.7 V. The diode is 28) C
A) conducting. B) shorted. C) reverse-biased. D) forward-biased.

29) A 6.2 V zener is rated at 1 watt. The maximum safe current the zener can carry is approximately 29) D
A) 1.61 A. B) 16.1 mA. C) 1.61 mA. D) 161 mA.

30) The knee voltage of a diode is approximately equal to the 30) B
A) reverse voltage. B) barrier potential.
C) applied voltage. D) breakdown voltage.

31) Refer to Figure XXX. Which of the oscilloscope trace indicates the output from a full wave rectifier with ide 31) B
diodes
A) b B) c C) a. D) d

32) When a zener diode is forward-biased, it 32) C
A) exhibits a voltage drop of V_Z + 0.7 V. B) conducts no current.
C) operates like a rectifier diode. D) emits visible light.

33) The dc current through each diode in a bridge rectifier equals 33) C
A) one-fourth the dc load current. B) twice the dc load current.
C) the load current. D) half the dc load current.

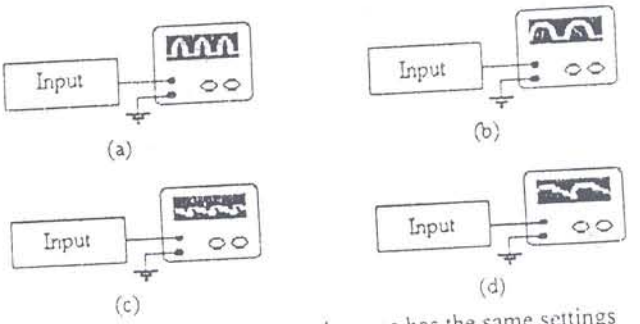


Figure XXX: Assume each scope has the same settings

34) Refer to Figure XXX. which oscilloscope trace indicates the output from a properly operating full-wave rectifi 34) A
with a filter?
A) c. B) d. C) a. D) b.

35) Refer to Figure XXX. Which oscilloscope trace indicates the output from a properly operating half-wave rect 35) A
without a filter
A) a. B) b. C) d. D) c.

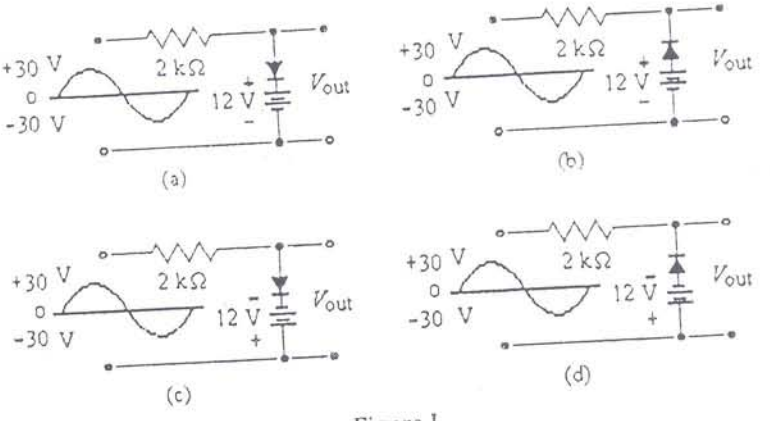


Figure I

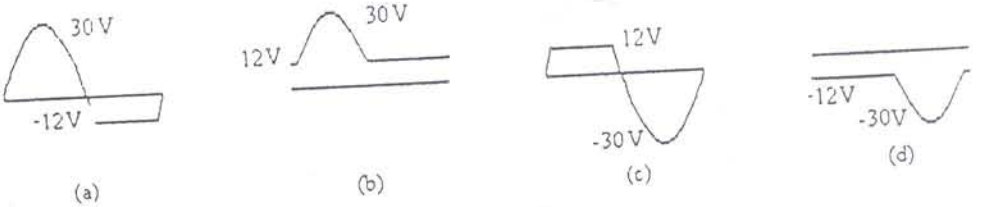


Figure II

36) B
36) Which of the circuits in Figure I will produce the signal in Figure II (b)?
A) (a) B) (b) C) (c) D) (d)

37) Which of the circuits in Figure I will produce the signal in Figure II (d)? 37) C
A) (a) B) (b) C) (c) D) (d)

38) Which of the circuits in Figure I will produce the signal in Figure II (a)? 38) D
A) (a) B) (b) C) (c) D) (d)

39) Which of the circuits in Figure I will produce the signal in Figure II (c)? 39) A
A) (a) B) (b) C) (c) D) (d)

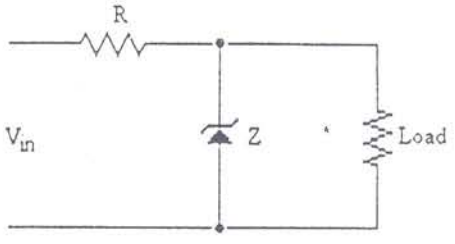


Figure Y

40) Refer to the Figure Y. If the zener short circuits, 40) D
A) V_{RL} will be equal to zero. B) there is no current through R_L.
C) V_{in} will be equal to V_R. D) All of the above.

41) Refer to the Figure Y. As long as the zener is in regulation, 41) D
A) there is no current through the zener. B) there is no current through R_L.
C) V_{RL} will be equal to 0.7 V. D) the current (I_{RL}) remains constant.

42) Refer to the Figure Y. If V_{in} increases, V_R will 42) B
A) remain the same. B) increase.
C) decrease. D) Unable to determine.

43) Refer to the Figure Y. Measurements show that R_L has increased. Which of the following faults, if any, could have caused this problem? 43) A
A) The zener opens. B) R opens.
C) V_{in} has decreased. D) The zener shorts.

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RUE/FALSE. Write 'T' if the statement is true and 'F' if the statement is false.

44) A free electron has a higher energy state than any that are bound to their nucleus. 44) T

45) The depletion layer of a p-n junction increases with increase in forward bias voltage 45) F

46) Electrons are the minority carriers in an n-type material. 46) T

47) The larger the ripple voltage, the better the filter. 47) F

48) Semiconductor material of the p-type has no free electrons. 48) F

49) In the depletion region of a p-n junction, the n-side is positively charged and the p-side is negatively charged. 49) F

50) Reverse bias permits full current through a pn junction. 50) F